HW#5

Homework Report

Course	ECG 762
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Problem 01 5.2

The IID observations $x[n]$ for $n = 0,1, \dots, N-1$ have the Rayleigh PDF

$$p(x[n];\sigma^2) = \begin{cases} \dfrac{x[n]}{\sigma^2}\exp\left(-\dfrac{1}{2}\dfrac{x^2[n]}{\sigma^2}\right) & x[n] > 0 \\ 0 & x[n] < 0. \end{cases}$$

Find a sufficient statistic for σ^2.

Solution:

Since the observations are independent, their joint PDF is

$$p(\mathbf{x};\sigma^2) = \prod_{n=0}^{N-1} \frac{x[n]}{\sigma^2}\exp\left(-\frac{x^2[n]}{2\sigma^2}\right), x[n] > 0 \text{ for all } n.$$

Rearranging gives

$$p(\mathbf{x};\sigma^2) = (\sigma^2)^{-N}\exp\left[-\frac{1}{2\sigma^2}\sum_{n=0}^{N-1} x^2[n]\right]\prod_{n=0}^{N-1} x[n].$$

Define

$$T(\mathbf{x}) = \sum_{n=0}^{N-1} x^2[n].$$

The joint PDF has the factorization

$$p(\mathbf{x};\sigma^2) = g(T(\mathbf{x}),\sigma^2)h(\mathbf{x}),$$

where

$$g(T,\sigma^2) = (\sigma^2)^{-N}\exp\left(-\frac{T}{2\sigma^2}\right), h(\mathbf{x}) = \prod_{n=0}^{N-1} x[n].$$

The support does not depend on σ^2, and $h(\mathbf{x})$ contains no unknown parameter.
Therefore, by the Neyman-Fisher factorization theorem, a sufficient statistic for σ^2 is

$$T(\mathbf{x}) = \sum_{n=0}^{N-1} x^2[n].$$

Problem 02 5.3
The IID observations $x[n]$ for $n = 0,1, \dots, N-1$ have the exponential PDF

$$p(x[n];\lambda) = \begin{cases} \lambda \exp(-\lambda x[n]) & x[n] > 0 \\ 0 & x[n] < 0. \end{cases}$$

Find a sufficient statistic for λ.

Solution:

The joint PDF of the N independent observations is

$$p(\mathbf{x};\lambda) = \prod_{n=0}^{N-1} \lambda e^{-\lambda x[n]}$$
$$= \lambda^N \exp\left[-\lambda \sum_{n=0}^{N-1} x[n]\right], x[n] > 0 \text{ for all } n.$$

Let

$$T(\mathbf{x}) = \sum_{n=0}^{N-1} x[n].$$

Then the joint PDF can be written as

$$p(\mathbf{x};\lambda) = g(T(\mathbf{x}),\lambda)h(\mathbf{x}),$$

with

$$g(T,\lambda) = \lambda^N e^{-\lambda T}, h(\mathbf{x}) = 1$$

on the parameter-independent support.

Thus, by the Neyman-Fisher factorization theorem, a sufficient statistic for λ is

$$T(\mathbf{x}) = \sum_{n=0}^{N-1} x[n].$$

The sample mean is also sufficient because it is a one-to-one transformation of $T(\mathbf{x})$ for a fixed N.

Problem 03 5.6

If $x[n] = A + w[n]$ for $n = 0,1,\ldots,N-1$ are observed, where $w[n]$ is WGN with variance σ^2, find the MVU estimator for σ^2 assuming that A is known. You may assume that the sufficient statistic is complete.

Solution:

Since A is known and the noise samples are independent Gaussian random variables,

$$x[n] \overset{\text{iid}}{\sim} \mathcal{N}(A,\sigma^2).$$

The joint PDF is therefore

$$p(\mathbf{x};\sigma^2) = (2\pi\sigma^2)^{-N/2}\exp\left[-\frac{1}{2\sigma^2}\sum_{n=0}^{N-1}(x[n]-A)^2\right].$$

By the Neyman-Fisher factorization theorem,

$$T(\mathbf{x}) = \sum_{n=0}^{N-1}(x[n]-A)^2$$

is sufficient for σ^2.

To obtain an unbiased estimator, note that

$$x[n]-A = w[n], E[w^2[n]] = \sigma^2.$$

Hence,

$$E[T(\mathbf{x})] = \sum_{n=0}^{N-1} E[w^2[n]] = N\sigma^2,$$

so

$$E\left[\frac{T(\mathbf{x})}{N}\right] = \sigma^2.$$

By the assumed completeness of $T(\mathbf{x})$, the Rao-Blackwell-Lehmann-Scheffé theorem implies that the MVU estimator is

$$\widehat{\sigma^2}_{\text{MVU}} = \frac{1}{N}\sum_{n=0}^{N-1}(x[n]-A)^2.$$

The denominator is N, rather than $N-1$, because the mean A is known and does not need to be estimated.

Furthermore,

$$\frac{T(\mathbf{x})}{\sigma^2} \sim \chi^2_N,$$

which gives

$$\text{Var}\left(\widehat{\sigma^2}_{\text{MVU}}\right) = \frac{2\sigma^4}{N}.$$

Problem 04 5.7
Consider the frequency estimation of a sinusoid embedded in WGN or

$$x[n] = \cos 2\pi f_0 n + w[n]\ n = 0,1,\ldots,N-1$$

where $w[n]$ is WGN of known variance σ^2. Show that it is not possible to find a sufficient statistic for the frequency f_0.

Solution:

Strictly speaking, the full observation vector is always sufficient. The intended conclusion is that the observations containing frequency information cannot be reduced to a simple summary statistic, such as a sample sum or a sum of squares, without losing information.

The joint PDF is

$$p(\mathbf{x}; f_0) = (2\pi\sigma^2)^{-N/2}\exp\left[-\frac{1}{2\sigma^2}\sum_{n=0}^{N-1}\left(x[n]-\cos(2\pi f_0 n)\right)^2\right].$$

Expanding the squared terms yields

$$p(\mathbf{x}; f_0) = h(\mathbf{x})\exp\left[\frac{1}{\sigma^2}\sum_{n=0}^{N-1}x[n]\cos(2\pi f_0 n) - \frac{1}{2\sigma^2}\sum_{n=0}^{N-1}\cos^2(2\pi f_0 n)\right],$$

where

$$h(\mathbf{x}) = (2\pi\sigma^2)^{-N/2}\exp\left[-\frac{1}{2\sigma^2}\sum_{n=0}^{N-1}x^2[n]\right]$$

does not depend on f_0.
The term coupling the data to the unknown frequency is

$$\sum_{n=0}^{N-1}x[n]\cos(2\pi f_0 n).$$

Its weights depend on f_0, so this expression cannot itself serve as a statistic: a statistic must be computable from the observations and known quantities alone.

To establish the absence of further information-preserving reduction, consider two observation vectors **x** and **y**. Their log-likelihood ratio is

$$\log\frac{p(\mathbf{x}; f_0)}{p(\mathbf{y}; f_0)} = C(\mathbf{x}, \mathbf{y}) + \frac{1}{\sigma^2}\sum_{n=1}^{N-1}(x[n]-y[n])\cos(2\pi f_0 n),$$

where $C(\mathbf{x}, \mathbf{y})$ is independent of f_0.
When the frequency parameter ranges over a nondegenerate interval, the functions

$$1, \cos(2\pi f_0), \ldots, \cos(2\pi(N-1)f_0)$$

are linearly independent. Consequently, the likelihood ratio is independent of f_0 if and only if

$$x[n] = y[n], n = 1, \ldots, N-1.$$

By the likelihood-ratio characterization of minimal sufficiency, a minimal sufficient statistic is therefore

$$T(\mathbf{x}) = (x[1], x[2], \ldots, x[N-1]).$$

The observation $x[0]$ can be discarded because

$$x[0] = 1 + w[0]$$

has a distribution independent of f_0 and is independent of the remaining observations. Thus, all frequency-bearing observations must be retained, up to an invertible transformation. The textbook's statement should be understood as ruling out a nontrivial reduction of these observations, rather than ruling out every sufficient statistic.

Problem 05 5.9
Assume that $x[n]$ is the result of a Bernoulli trial (a coin toss) with

$$\Pr\{x[n] = 1\} = \theta$$
$$\Pr\{x[n] = 0\} = 1 - \theta$$

and that N IID observations have been made. Assuming the Neyman-Fisher factorization theorem holds for discrete random variables, find a sufficient statistic for θ. Then, assuming completeness, find the MVU estimator of θ.

Solution:

For $x[n] \in \{0,1\}$, the Bernoulli probability mass function is

$$p(x[n]; \theta) = \theta^{x[n]}(1-\theta)^{1-x[n]}.$$

Since the observations are independent, their joint probability mass function is

$$\begin{aligned} p(\mathbf{x}; \theta) &= \prod_{n=0}^{N-1} \theta^{x[n]}(1-\theta)^{1-x[n]} \\ &= \theta^{\sum_{n=0}^{N-1} x[n]}(1-\theta)^{N-\sum_{n=0}^{N-1} x[n]}. \end{aligned}$$

Define

$$T(\mathbf{x}) = \sum_{n=0}^{N-1} x[n].$$

Then

$$p(\mathbf{x}; \theta) = \theta^{T(\mathbf{x})}(1-\theta)^{N-T(\mathbf{x})} \cdot 1.$$

By the Neyman-Fisher factorization theorem, the total number of successes,

$$T(\mathbf{x}) = \sum_{n=0}^{N-1} x[n],$$

is sufficient for θ.

To obtain an unbiased estimator, use $E[x[n]] = \theta$:

$$E[T(\mathbf{x})] = \sum_{n=0}^{N-1} E[x[n]] = N\theta.$$

Therefore,

$$E\left[\frac{T(\mathbf{x})}{N}\right] = \theta.$$

Since $T(\mathbf{x})$ is assumed complete, the Rao-Blackwell-Lehmann-Scheffé theorem gives

$$\hat{\theta}_{\text{MVU}} = \frac{1}{N}\sum_{n=0}^{N-1} x[n].$$

Thus, the MVU estimator is the observed proportion of successes. Its variance is

$$\text{Var}\left(\hat{\theta}_{\text{MVU}}\right) = \frac{1}{N^2}\sum_{n=0}^{N-1} \theta(1-\theta) = \frac{\theta(1-\theta)}{N}.$$